

Cancer is a disease of the genome. Something goes wrong in a cell's DNA, that cell stops following normal rules about when to grow and when to die, and it multiplies out of control. Reading the genetic code of a tumor is expensive and requires equipment no high school has, and getting tumor samples requires hospital partnerships, ethics approval, and years of work. The Cancer Genome Atlas solves that for you. Run by the National Cancer Institute and the National Human Genome Research Institute, it molecularly characterized over 20,000 primary cancer and matched normal samples spanning 33 cancer types, generated more than 2.5 petabytes of data, and made all of it publicly available to anyone. It covers over 11,000 patients, with between 50 and 1,500 patients sampled per tumor type. Everyone calls it TCGA.

Across the patients, they collected multiple layers of data. That includes clinical information about the patient, metadata about the tissue samples, histopathology slide images of the tumor tissue, and molecular data covering mRNA expression, microRNA expression, protein expression, and copy number changes. Mutation data, gene expression, methylation, and clinical records all sit together, including survival outcomes. Because they recorded how long patients lived and whether they died, you can connect a molecular measurement to a real human outcome.

Four main data types:

- Gene expression tells you how active a gene is in that tumor, meaning how much the cell is actually using those instructions. A gene can be present in the DNA but barely switched on, or activated far beyond normal, and cancer often involves genes being expressed at the wrong level.
- Mutation data tells you whether the DNA sequence itself is altered, which is a separate question from how active it is.
- Copy number tells you whether the tumor has extra or missing copies of a chunk of DNA, since cancer cells frequently duplicate or delete whole regions.
- Clinical data is the patient information: age, tumor stage, treatment, and critically, how long they survived and whether they're still alive.

Your project needs to start with a question specific enough to test. "Does this gene cause cancer" is unanswerable. **"In lung adenocarcinoma patients, do those with high expression of gene X survive longer than those with low expression" is answerable.** Good question shapes include whether expression of a gene predicts survival within one cancer type, whether patients carrying a mutation have worse outcomes than those without it, whether a gene known to matter in one cancer behaves the same way in another, and whether two genes in the same biological pathway show correlated expression in real tumors.

Start with UCSC Xena, which runs in your browser at xenabrowser.net and needs no coding. It does Kaplan-Meier survival analysis, filtering and subgrouping, statistical analysis, box plots, and scatter plots, and exports your figure as a PDF. A Kaplan-Meier curve is the standard way survival is shown in medicine: time runs along the bottom, percentage of patients still alive runs up the side, and the line steps down each time someone dies. Split patients into high-expression and low-expression groups, then use the log-rank test to determine whether the

difference between the curves is statistically significant. Xena lets you pick which survival endpoint you're measuring, like overall survival versus recurrence-free survival, and set a custom cutoff such as ten years. A published example from the Xena team shows patients with the astrocytoma subtype of lower grade glioma had significantly worse ten-year survival than those with oligodendroglioma or oligoastrocytoma subtypes.

The other main tool is cBioPortal at cbioportal.org, which is better for mutations and alterations. It's an open source site for interactive exploration, visualization, and download of cancer genomics datasets. Published papers spell out the workflow step by step: select a cancer study, choose your genomic profile such as mRNA expression z-scores, enter your gene, submit, then open the Survival tab to get the Kaplan-Meier estimate for overall survival. You'll also get an OncoPrint, a visual grid showing which patients have alterations in your gene and what kind. Standard published workflows use Kaplan-Meier analysis with a log-rank test to compare survival between patients with and without an alteration. The log-rank test tells you whether the gap between two curves is real or noise, and produces a p-value, with below 0.05 the conventional threshold for significance. A significant result shows an association between the gene and survival, not that the gene directly causes differences in patient outcomes.

Before you commit, search Google Scholar and PubMed for your gene plus your cancer type. If forty papers already cover it, pick something else or find the angle they skipped. Good project angles include cross-cancer comparisons, since a gene studied heavily in breast cancer may be nearly unexamined in a rarer type, and subgroup analyses, where a finding that holds overall might not hold in younger patients or a particular molecular subtype. Validation is also legitimate, meaning you take a published claim and test whether it holds in a TCGA cohort the original authors didn't use.

A realistic first project:

Pick one cancer type with a decent patient count, pick one gene you have a biological reason to care about from reading a review paper on that cancer, open Xena or cBioPortal, split patients by expression at the median, generate the Kaplan-Meier curve, and record the number of patients in each group and the log-rank p-value. Then repeat in a second and third cancer type to see whether the pattern holds or breaks. Write up everything including the negative results, because a gene that predicts survival in one cancer and not another is more interesting than a single positive hit. To escalate, move to R, pull data with TCGAbiolinks, run a Cox model controlling for age and stage, and screen a set of related genes with proper multiple testing correction.